

Practical No. 02
ZOO 1203 - General Entomology
Insect Collection, preservation and mounting

Objectives:

After completing the practical you should be able to;

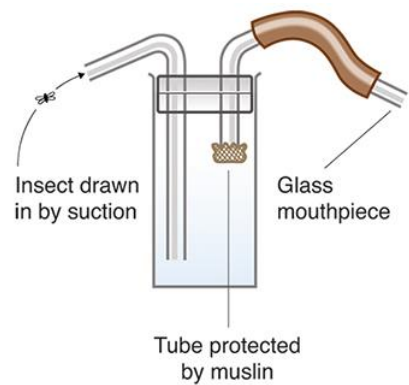
- Collect insects by using appropriate method and preserve them
 - Prepare them for mounting
 - Label the insects properly and make insect lab specimens
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Collecting and preservation

Collecting Equipment;

Aspirator
Beating sheet
Berlese funnel
Flight Intercept Trap
Light trap
Malaise trap
Owen Emergence Trap
Pitfall trap

1. Insect are usually caught by sweeping trees and bushes with an insect net or beaten from the vegetation onto a tray. They can also be picked up from underneath logs or rocks and are sometimes found in rotting wood. Aquatic insects can be caught with a net, normally a special heavy duty one.
2. An aspirator is used to get small insects out of the net or off the beating tray without damaging them. This is done by sucking on the short tube and placing the longer one over the insect.
3. The insects are then transferred into a killing bottle and left for at least one hour. Insects can also be killed by placing in freezer for 2 hours or more.



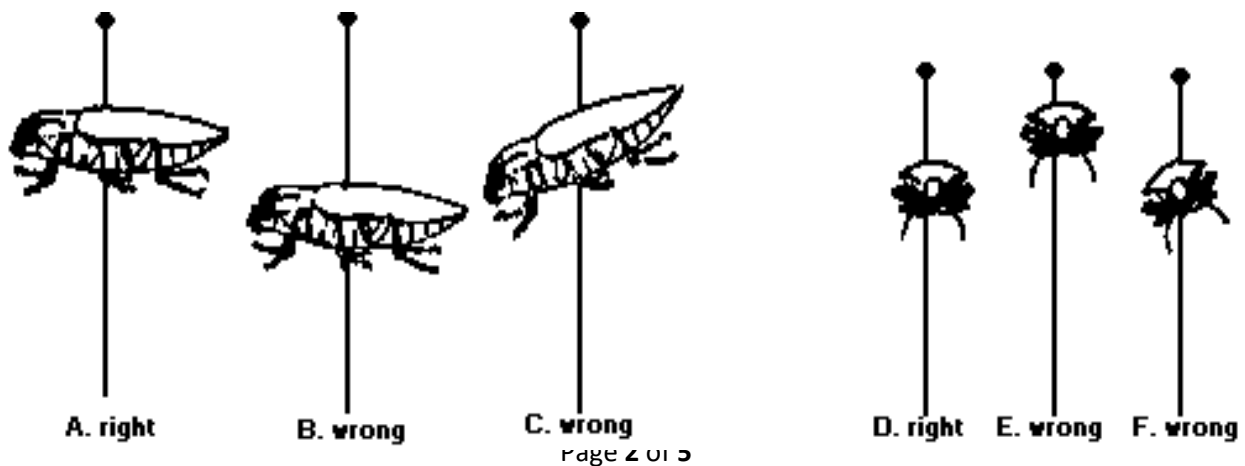
Aspirator

4. Most adult insect are best preserved dry by pinning. Immature eggs are best preserved in liquid. Very small insects can be preserved mounting on microscopic slides.
5. For liquid preservation place insect directly into 70% alcohol.
6. Large (longer than 2.5 cm) insect larvae and other soft bodies specimens should be fixed in Carny's Fluid or P.E.A. for 24 hours then transferred to alcohol.
7. If the specimens caught are not to be store in liquid preservative they should be mounted with 24 hours after capture while they are still soft and flexible.
8. If insect have to be kept in dry storage for more than a day they can be relaxed again by putting them in a dish on a piece of damp blotting paper in an air-tight container. They should not be left for more than a day as under the damp conditions mould will very rapidly grow on them.

Mounting of Insect

1. Stainles steel N0.3 pins are used for putting through the right hand side of insects larger than about 8 mm. Smaller insects are double mounted; in this case Black lacquered No.3 pins are to hold the double mounts. Black no.1 pins may be used for holding insect appendages in place when setting insect on a board.
2. Coleoptera (beetles) and Hemiptera (bugs) are mounted by placing a stainless steel pin through the right hand side of the abdomen.
3. Diptera (flies), Lepidoptera (moths and butterflies), Hymenoptera (bees and wasps) and Orthoptera (Grasshppers) are mounted by placing a stainless steel pin through the thorax.

Correct and incorrec pinned insects



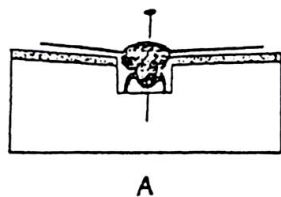
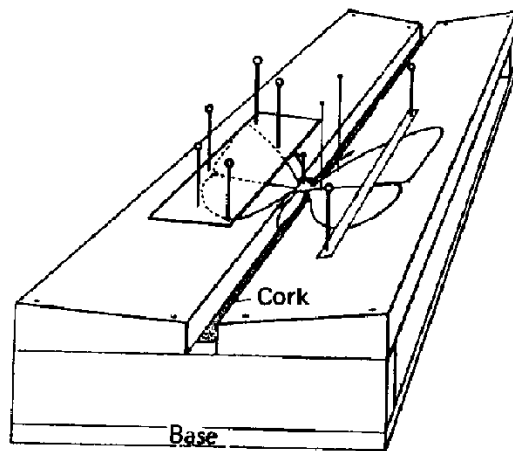
Lepidoptera and Orthoptera

They are mounted on boards with a longitudinal groove down the centre.

Grasp the butterfly by its thorax (middle of the body) with forceps, keeping the edge of the forceps just below the wing joint.

Choose a groove size that comfortably fits the width of the butterfly. (If the groove is too big, the butterfly will swivel around too much when you try to position the wings.)

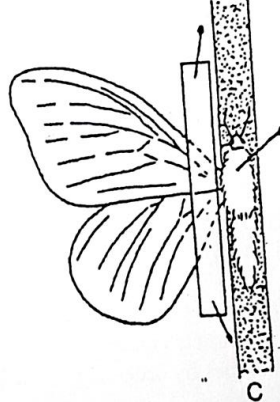
Then wings are spread and paper strips pinned down at both end.



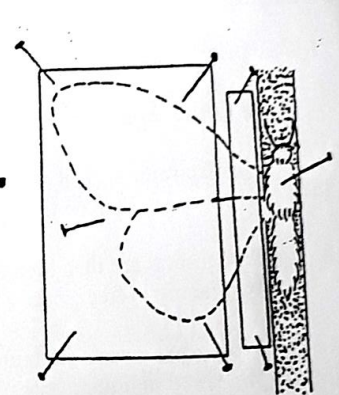
A



B



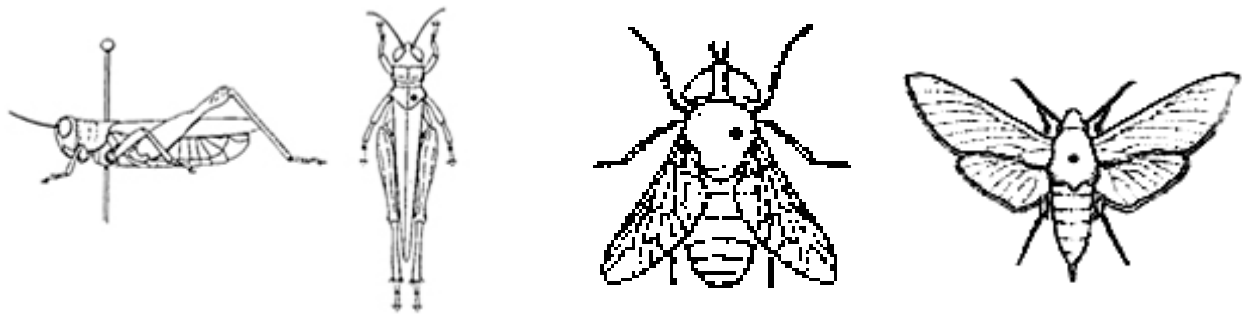
C



D

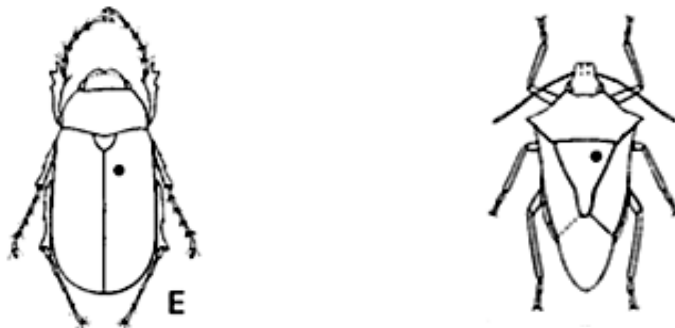
Orthoptera(grasshoppers, etc.), Hymenoptera(bees and wasps), Diptera(flies)

Are mounted by placing a stainless steel pin through the RIGHT hand side of the THORAX.



Coleoptera(beetles), Hemiptera(bugs)

Are mounted by placing a stainless steel pin through the RIGHT hand side of the ELYTRA.

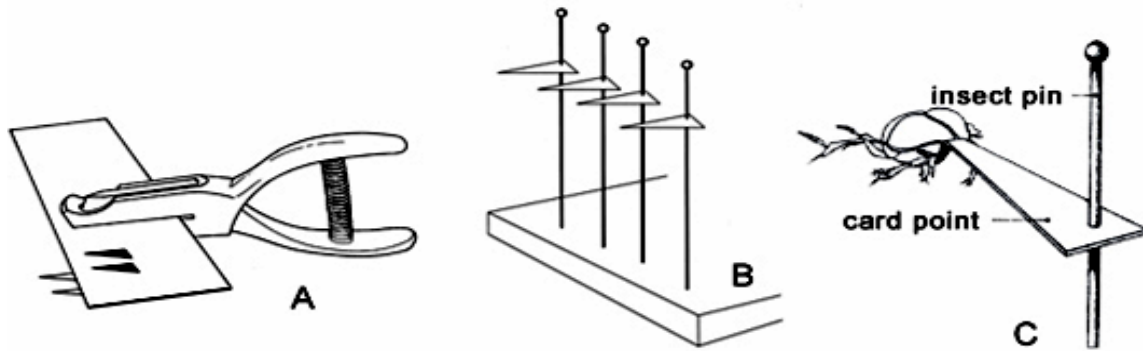


Double mounting of small insects

Triangle made from white card are used to mount very small Hemiptera, Coleoptera, Hymenoptera.

First small drop of glue placed on the end.

The insect is then glued to this on the right hand side of its thorax.



Labeling

Black label with ink using a very fine (0.1) fiber tipped pen or with Indian ink or a sharp pencil is acceptable, never use ballpoint. Also laser printer using 4 point type.

Labels should not exceed 18 x 12 mm.

Place labels on pin under insect or double mount.

For printed labels use up to 5 lines of text using the same Arial Narrow font at a size of 4 point. leave a blank line between labels.

Vial labels follow the same scheme as for pinned specimens, but using 4 lines of text and a larger point size (6).

